UNIVERSITY OF BIRMINGHAM

Quality-Controlled DIMS-Based Metabolomics

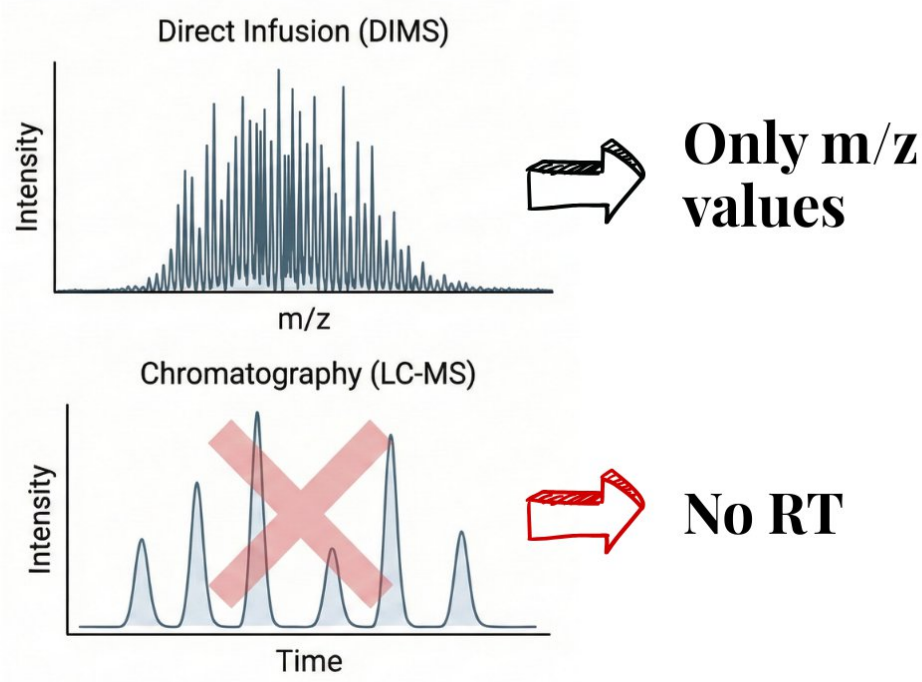
from Peak Detection to Putative Metabolite Annotation

Presented by Hattim, Ranveer, Shalini & Divya (Group 3)

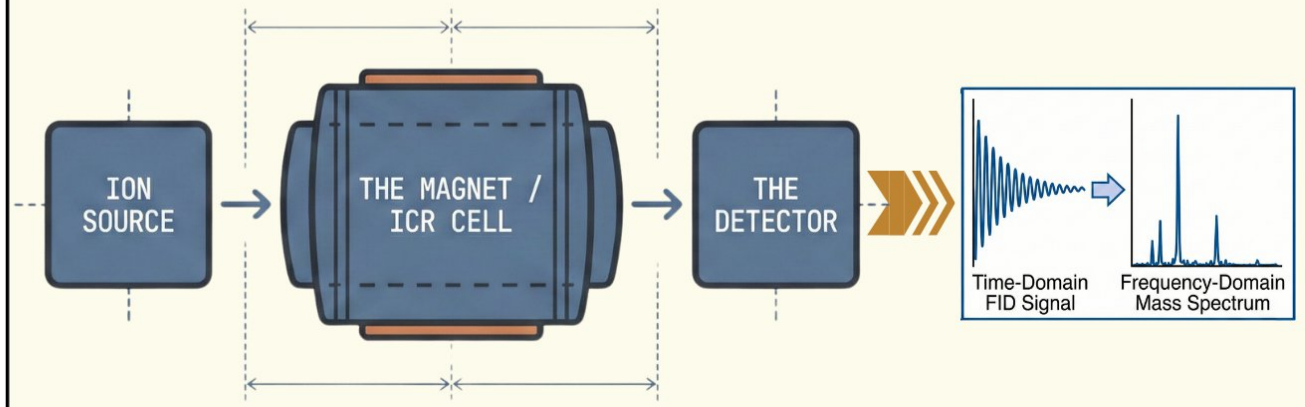


Brief Background

- ✓ Direct infusion mass spectrometry (DIMS)
- ✗ No chromatographic separation
- ✗ No MS/MS fragmentation



- **Instrument:** nESI Direct-infusion FT-ICR mass spectrometry (DIMS)
- **Mode:** Positive ion mode
- **Spectra acquisition method:** SIM-sticthing
- **Spectral Parameter:**
  - **mass Range:** ~70 - 590 Da
  - **No. of windows:** 7
  - **Window size:** 100 Da
  - **Overlap by:** 30 Da



Kirwan et al.

Aim & Objectives

Aim: To process and interpret DIMS data using established bioinformatics pipelines to identify reproducible metabolic features.



1. Transformation: Convert raw mzML files to normalized peak intensity matrix.



2. Sanitization: Apply rigorous filtering (replicate, blank, sample) to remove noise.



3. Exploration: Uncover global patterns via multivariate analysis (PCA).



4. Identification: Putative annotation using accurate mass and adduct patterns.



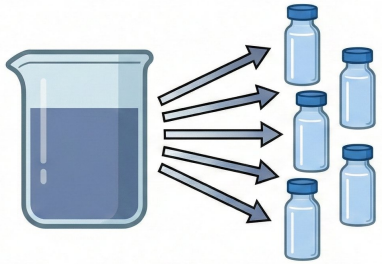
5. Context: Link metabolites to biological pathways.

Dataset Overview

STUDY CONTEXT

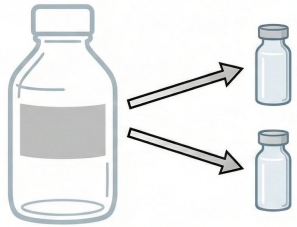
- Day 1 — Part of a multi-batch reproducibility study (8 Batches / 7 Days)
- Day 2
- Day 3 — **Batch 5** Acquired: Day 2
- Day 4 — **Significance:** Captures inter-day variation and technical drift following instrument adjustments.
- Day 5
- Day 6
- Day 7

Quality Control (QC)



- **Count:** 5 QC injections.
- **Source:** Pooled biological extracts.
- **Purpose:** Monitor technical stability and enable signal correction.

Blanks



- **Count:** 2 Blanks.
- **Source:** Solvent only.
- **Purpose:** Identify background contaminants for blank filtering.

Biological Samples



10 Cow Samples (Cardiac Tissue)
10 Sheep Samples (Cardiac Tissue)

20
Total Samples

Quality Control



5 QC Samples (Pooled Extracts)
2 Solvent Blanks

Purpose: Stability Monitoring & Contaminant Filtering

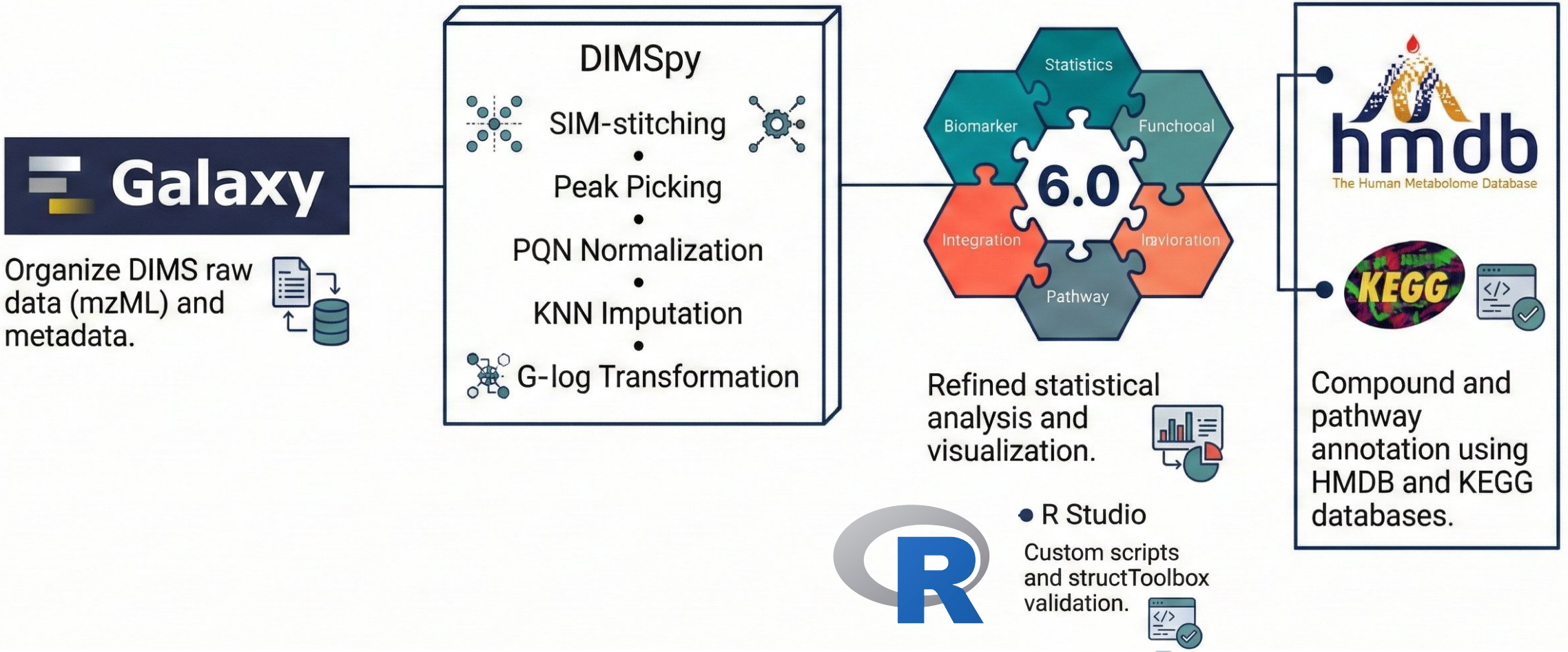
Method Rigor



Technical Triplicate Measurement

Purpose: Improving analytical reliability, and allowing peaks to be retained only if detected in at least two out of three runs, thereby reducing random noise and electrospray artefacts

Integrated Workflow

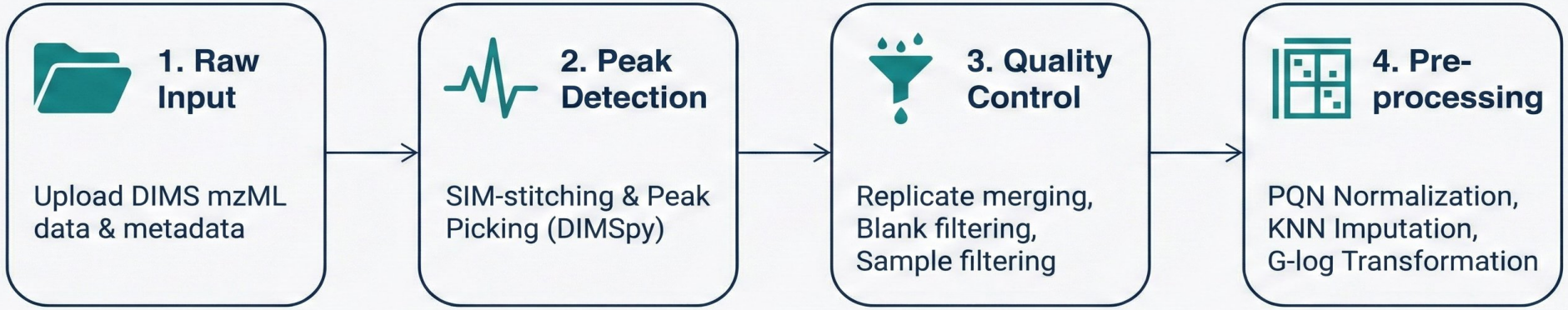


Data Processing



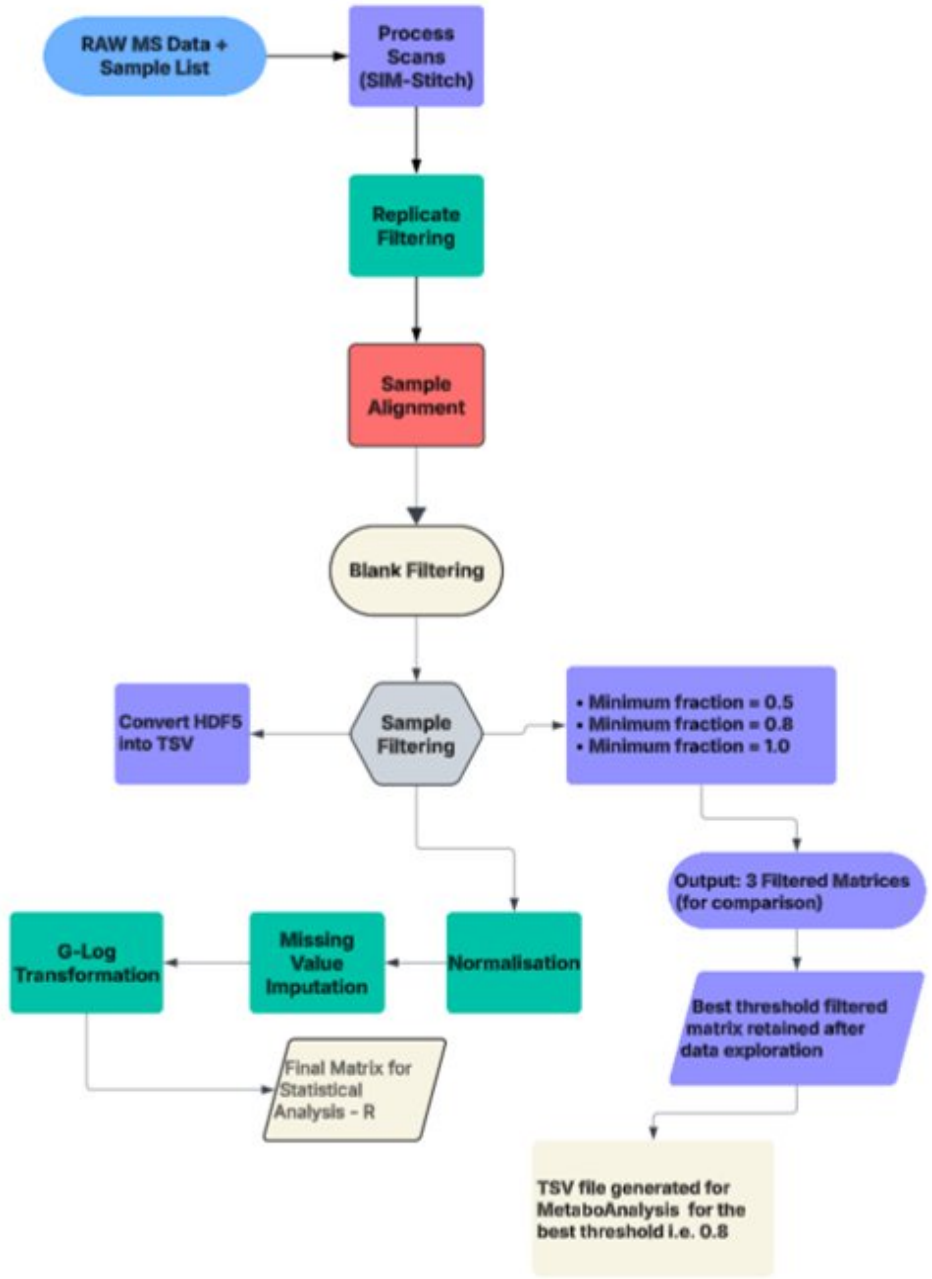
Galaxy – Processing tool

Data Processing Pipeline via Galaxy and DIMSpy



This workflow converts raw DIMS data into a clean, normalised matrix ready for statistical analysis

PIPELINE FOLLOWED



Used Parameters

Raw MS Data + Sample List
*.mzML files
Filelist / Sample metadata

1. Process Scans (SIM-Stitch)
Peak Detection & Stitching
Noise estimation (Median)
S/N ≥ 3
ppm tolerance = 2
Output: **Peaklist (HDF5)**

2. Replicate Filtering
Technical Replicate Filtering
3 replicates
ppm tolerance = 2
Output: **Filtered Peaklist**

3. Sample Alignment
Align Peaks Across Samples
ppm tolerance = 2
Output: **Peak Intensity Matrix (HDF5)**

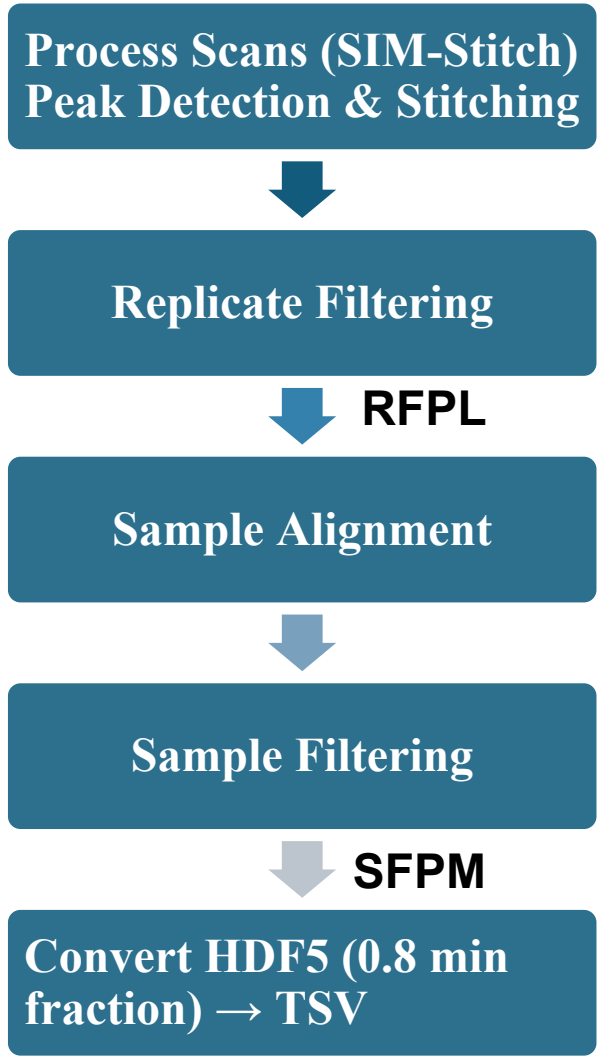
4. Sample Filtering
We tested 3 thresholds:
- Minimum fraction = 0.5
- Minimum fraction = 0.8
- Minimum fraction = 1.0

Output: **3 Filtered Matrices (for comparison)**

Best threshold filtered matrix retained after data exploration i.e. 0.8

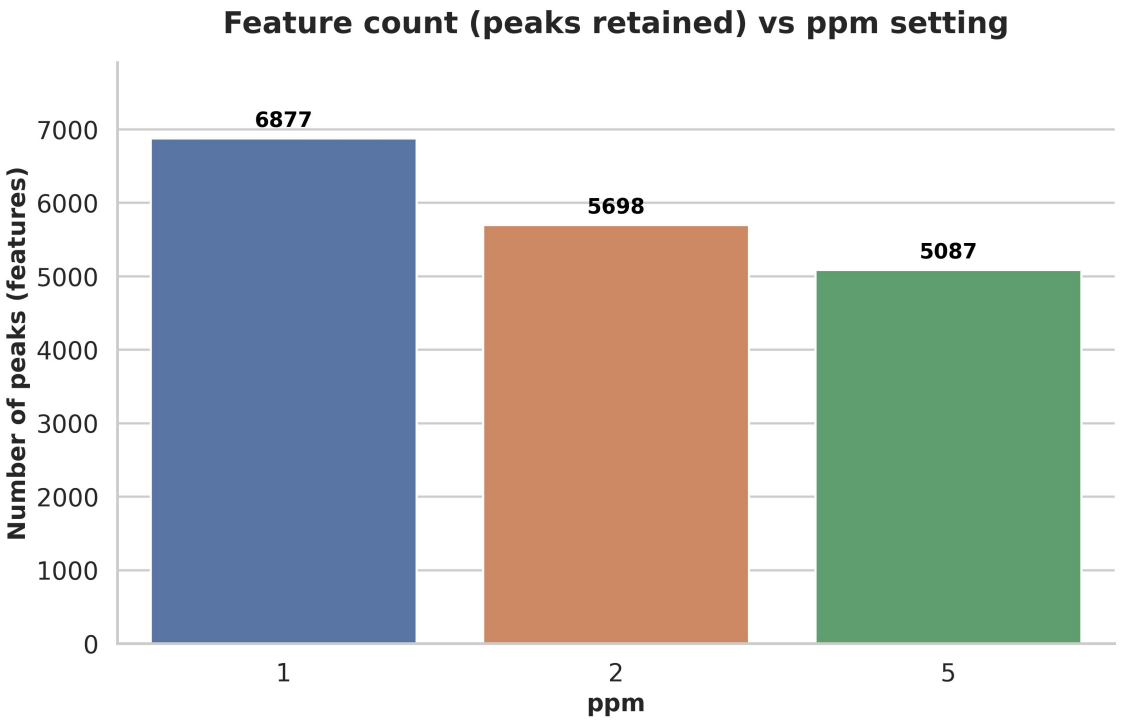
5. Convert HDF5 (0.8 min fraction) → TSV
Export Peak Intensity Matrix
- Rows = Samples
- Values = Intensity

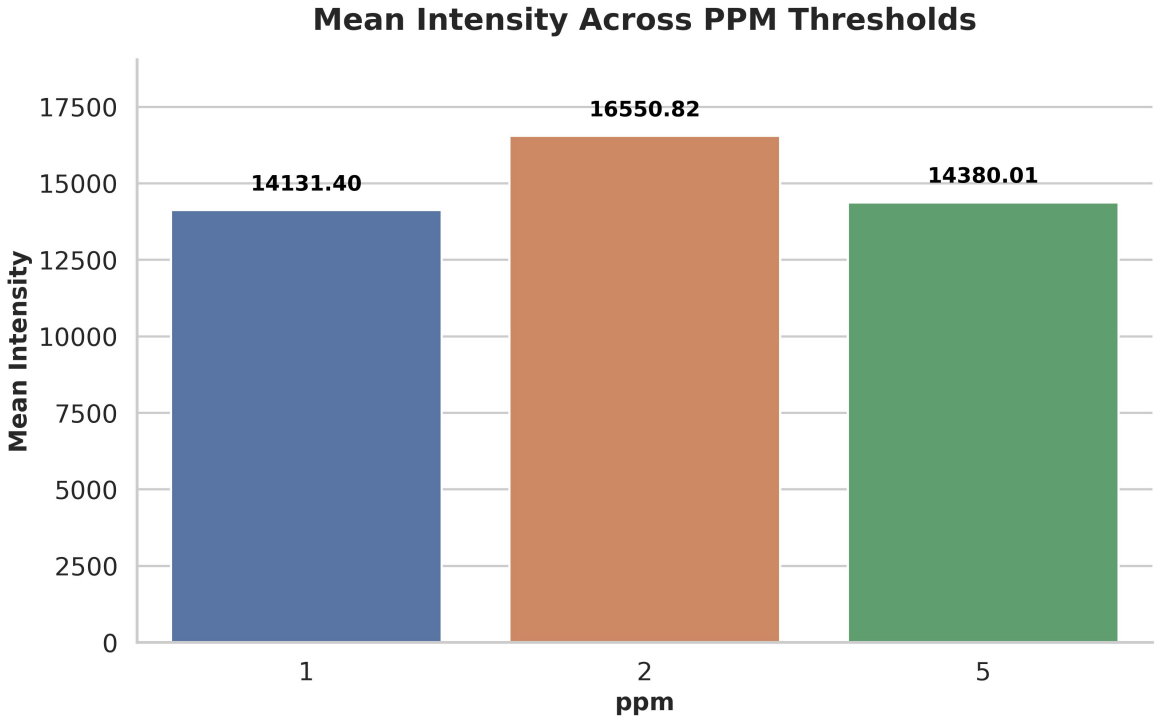
6. Data Refinement
- PQN Normalisation (QC-based)
- KNN Imputation (k=5)
- G-log transformation



Exploratory Data Analysis

PPM threshold comparison



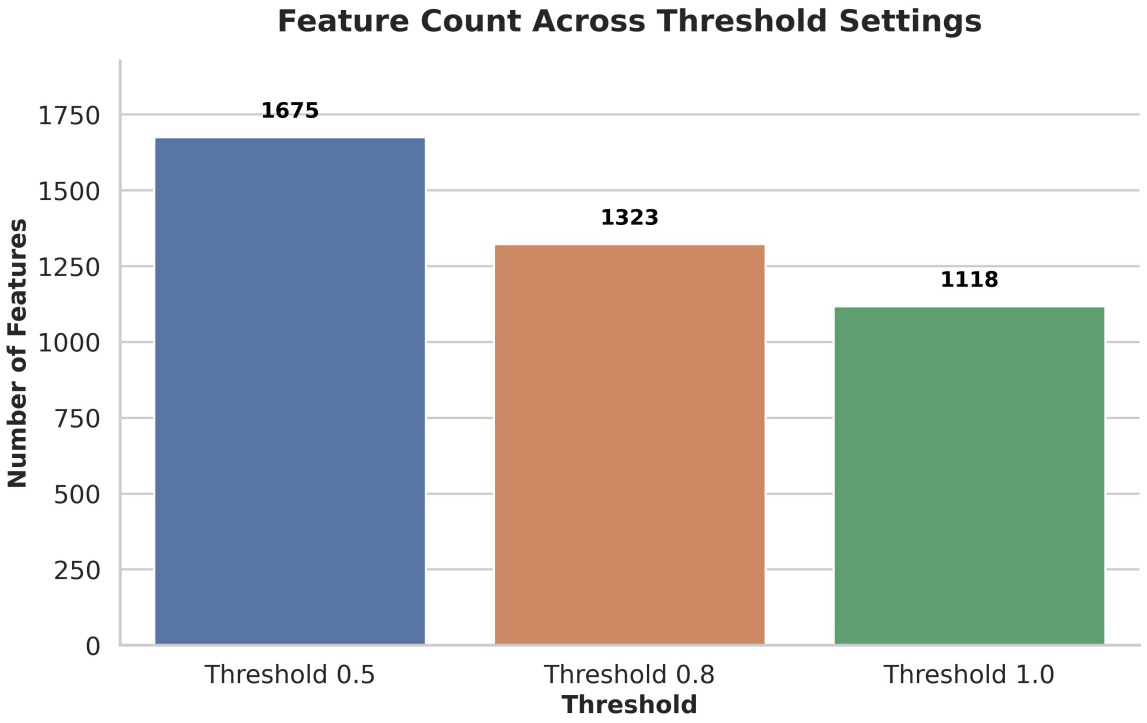


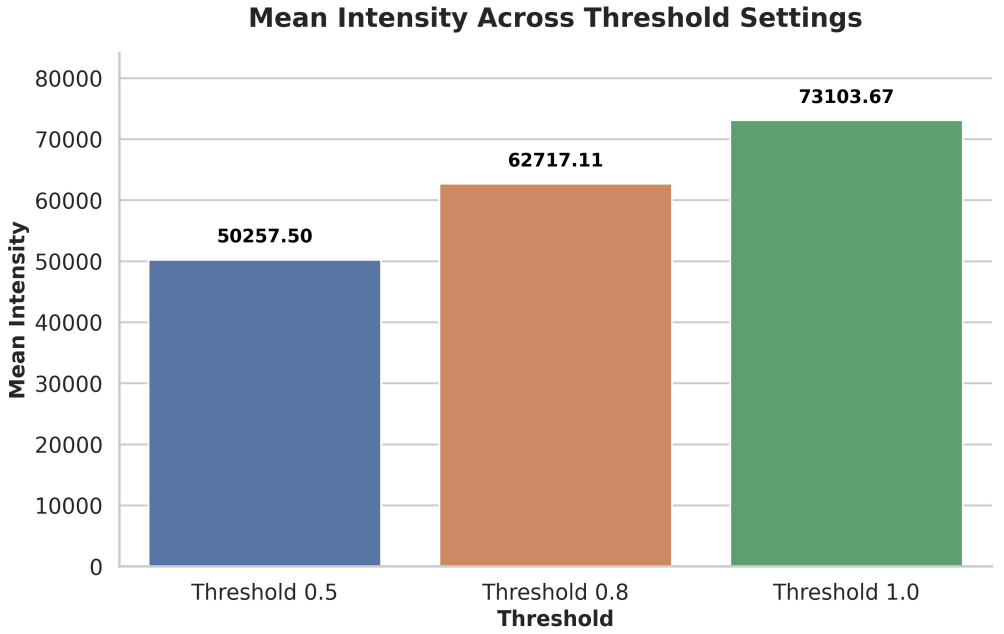
- **2 ppm** provides the optimal balance
- **1 ppm** threshold appears too strict, leading to artificial feature splitting.
- **5 ppm** threshold may be too permissive, increasing the risk of merging distinct signals.

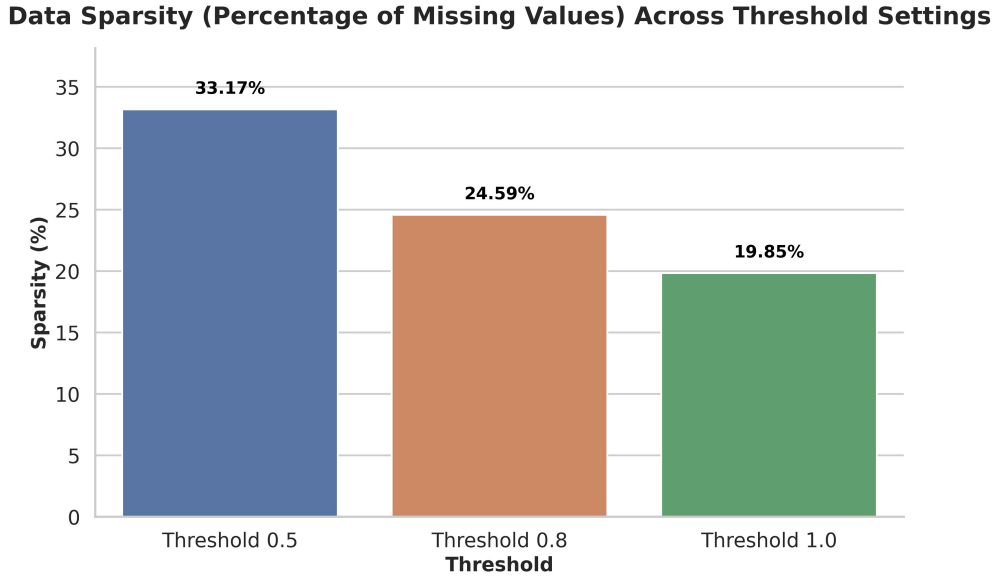
The 2 ppm setting reduces over-splitting, maintains appropriate resolution, and produces the strongest consolidated signal intensity, making it the most appropriate choice for high-resolution DIMS data.

Exploratory Data Analysis

Sample filter threshold comparison







Putting these three metrics together, we can see a clear trade-off.

- **0.5 threshold:** Most features but higher sparsity, so potentially more noise
- **1.0 threshold:** Least sparse dataset, but overly stringent and may remove biologically meaningful, class-specific features.
- **0.8 threshold:** Provides optimal balance

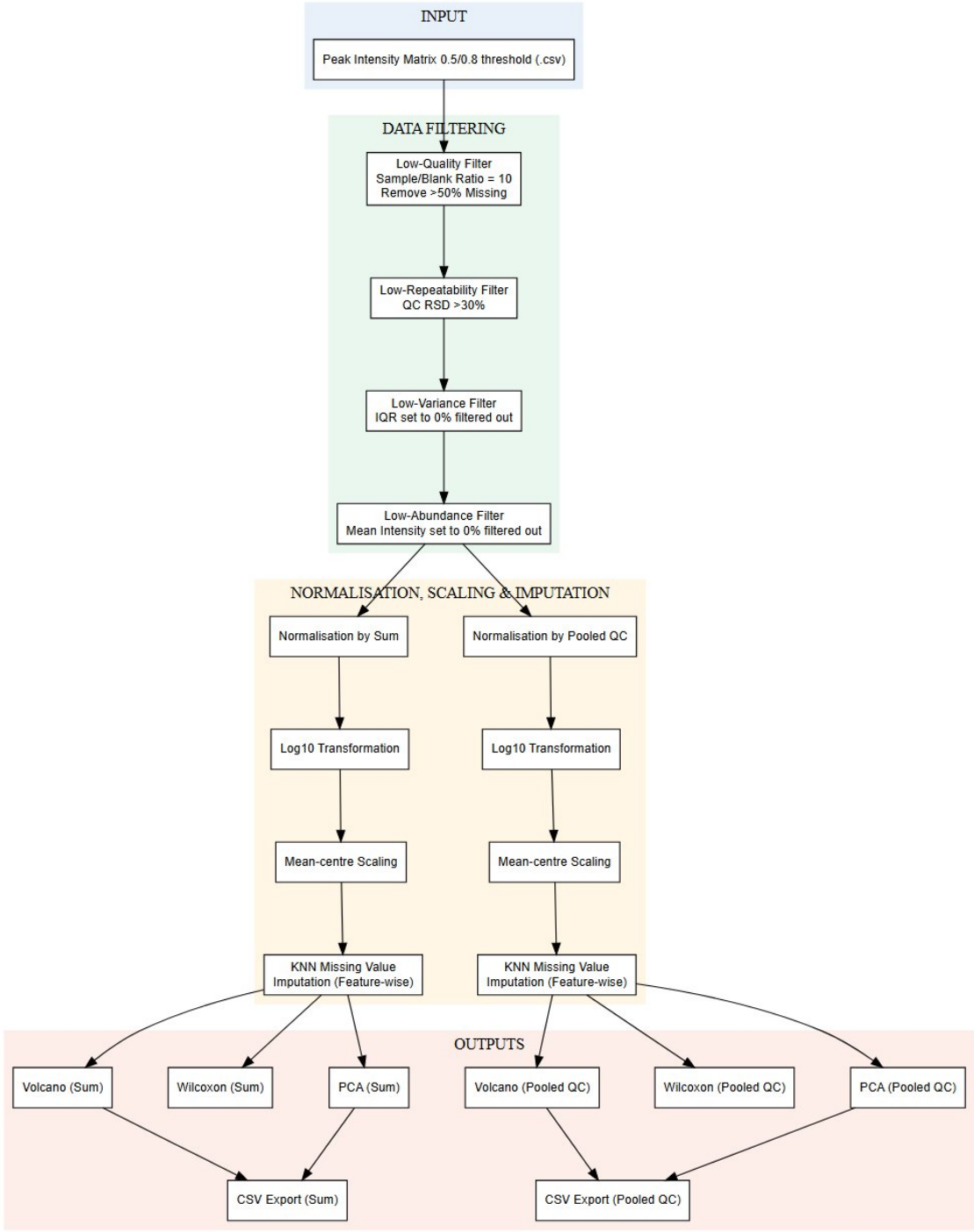
Therefore, a minimum fraction threshold of 0.8 is the most appropriate choice for downstream statistical analysis.

Data Analysis



MetaboAnalyst

Followed Pipeline

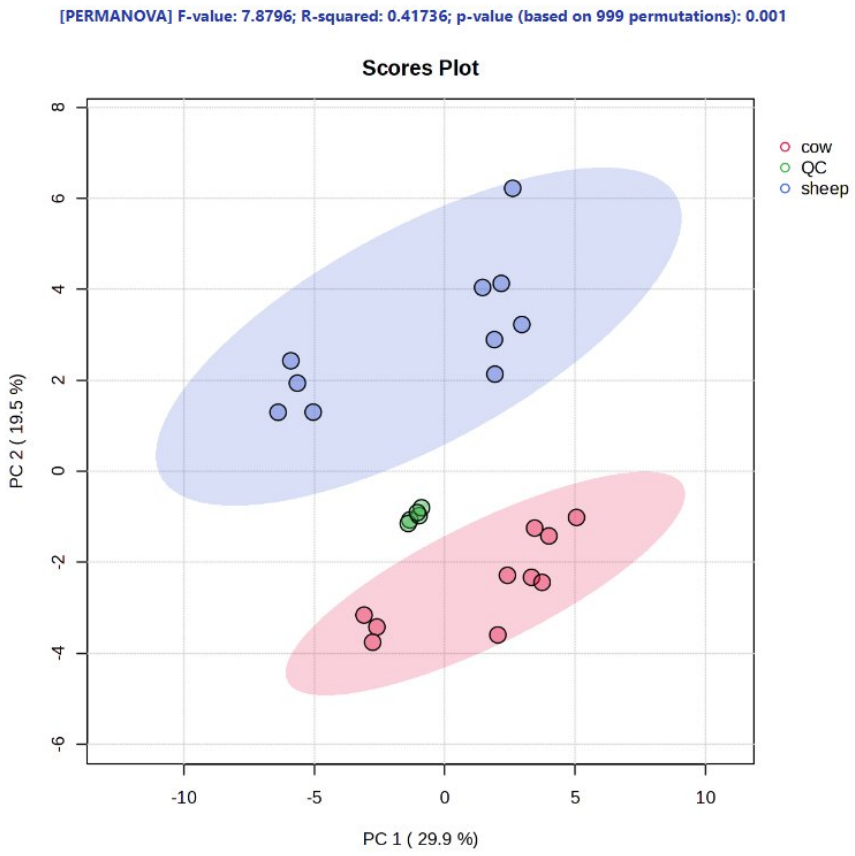


Justification of parameters

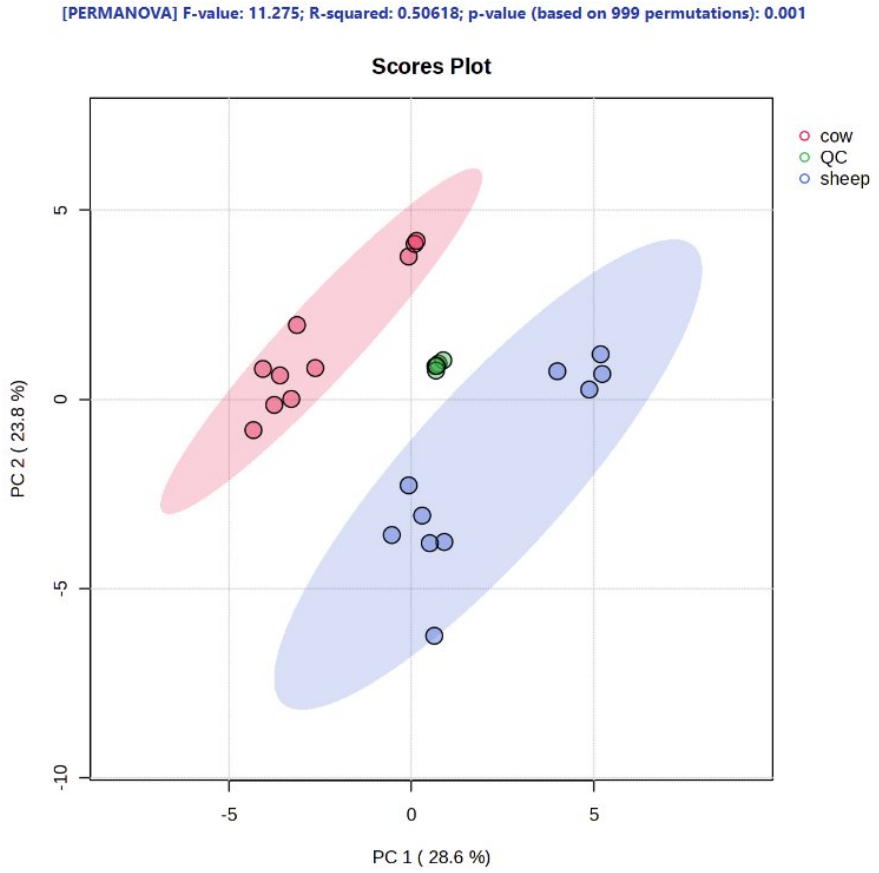
Parameter	Choice	Reasoning
Low-quality filter	Sample/blank ratio = 10 Remove features with >50% missing	Improves data quality by filtering out data with more than 50% missing values and a high SNR (indicates noise).
Low-repeatability filter	QC RSD >30%	Low RSD data are kept as they are more consistent and reliable, reducing variability.
Low-variance filter	IQR with 0% filtered out	Keeps all data to ensure no crucial information is removed for a comprehensive analysis.
Low-abundance filter	Mean intensity with 0% filtered out	Keeps low abundance data in the case of weak intensity e.g. rare metabolites present in the sample.
Normalisation by sum	-	Keeps the sum of values across samples constant, allowing for unbiased comparisons.
Normalisation by QC	-	Ensures consistency of the samples relative to the QC, reducing systematic variation.
Log10 transformation	-	Compresses large values and amplifies smaller ones, stabilising variance.
Mean-centre scaling	-	Centres the data around 0 by subtracting the mean, eliminating bias caused by the different magnitude of the features.
KNN (feature-wise)	-	Imputes missing values accurately using metabolites with similar m/z peaks for analysis.

Comparison of 0.5 and 0.8 threshold: PC1 vs PC2 Plot

0.5 threshold



0.8 threshold



- The F-value and R-squared value for 0.8 threshold is higher, suggesting it contains and explains more variance between the groups.
- There is overlapping between the points for each group for 0.8 threshold but overall, the separation is more distinct.
- The 95% confidence regions contain all the samples, but it is tighter with 0.8 threshold. This suggests that the 0.8 threshold dataset has less noise and is more precise.

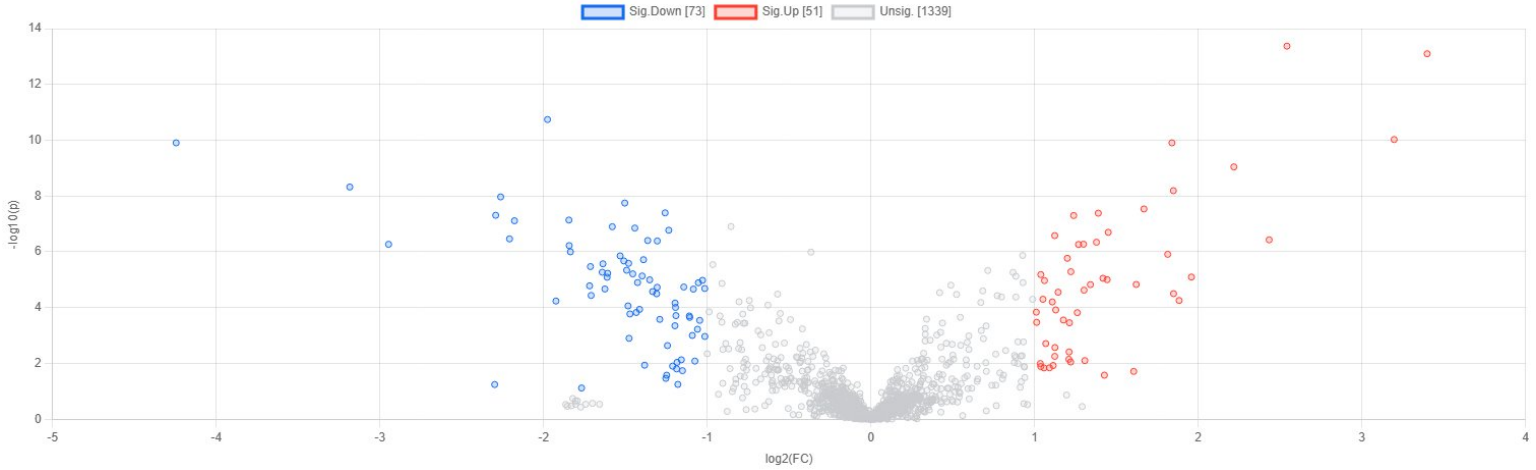
Comparison of 0.5 and 0.8 thresholds: Volcano plots

Upregulated metabolite count:

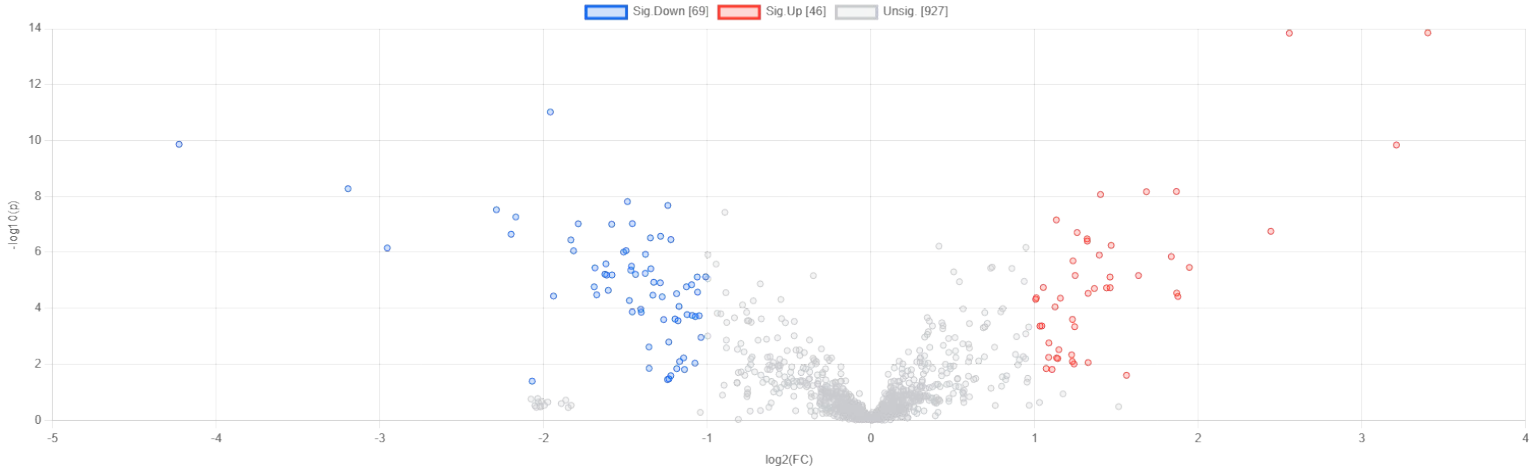
Dataset	Cow	Sheep	Insignificant
0.5 threshold	51	73	1339
0.8 threshold	69	46	927

- Sheep mean was the reference for the fold change.
- Increasing the threshold to 0.8 removed 421 low-quality/noise features.
- 0.8 threshold data contains fewer but more reliable significant points, resulting in a more distinct separation between features.
- In the 0.8 threshold volcano plot, there is a clearer distinction between the significant and non-significant points, making it easier to use for analysis.

0.5 threshold (FC threshold- 2, P-value threshold (Raw)- 0.1)

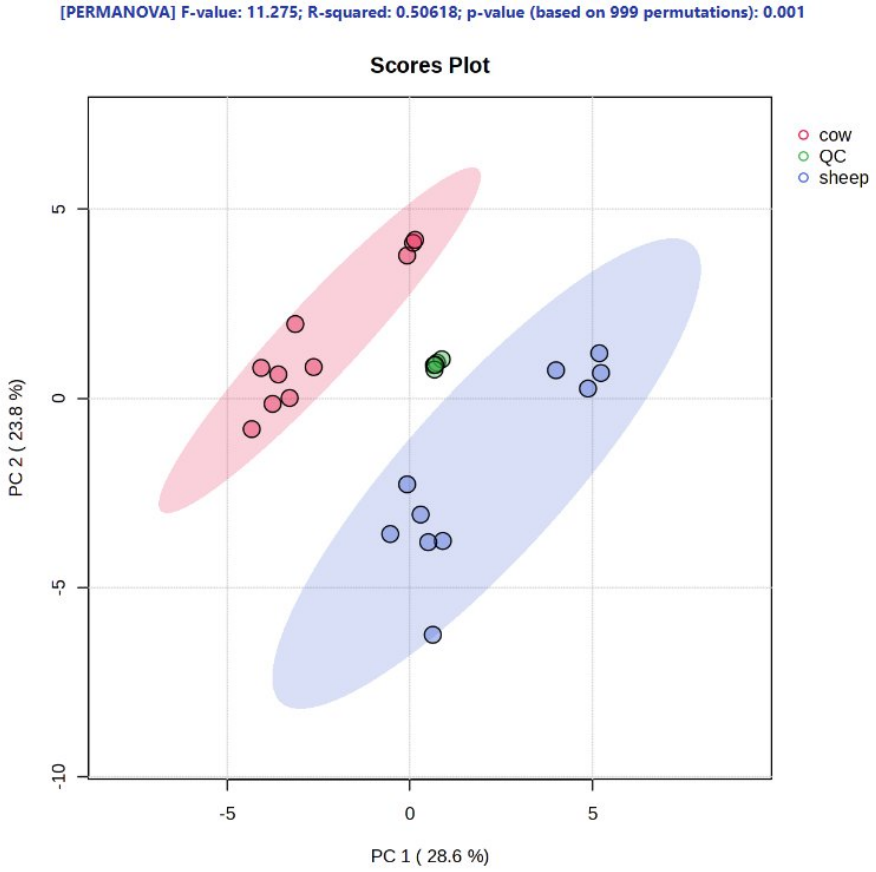


0.8 threshold (FC threshold- 2, P-value threshold (Raw)- 0.1)

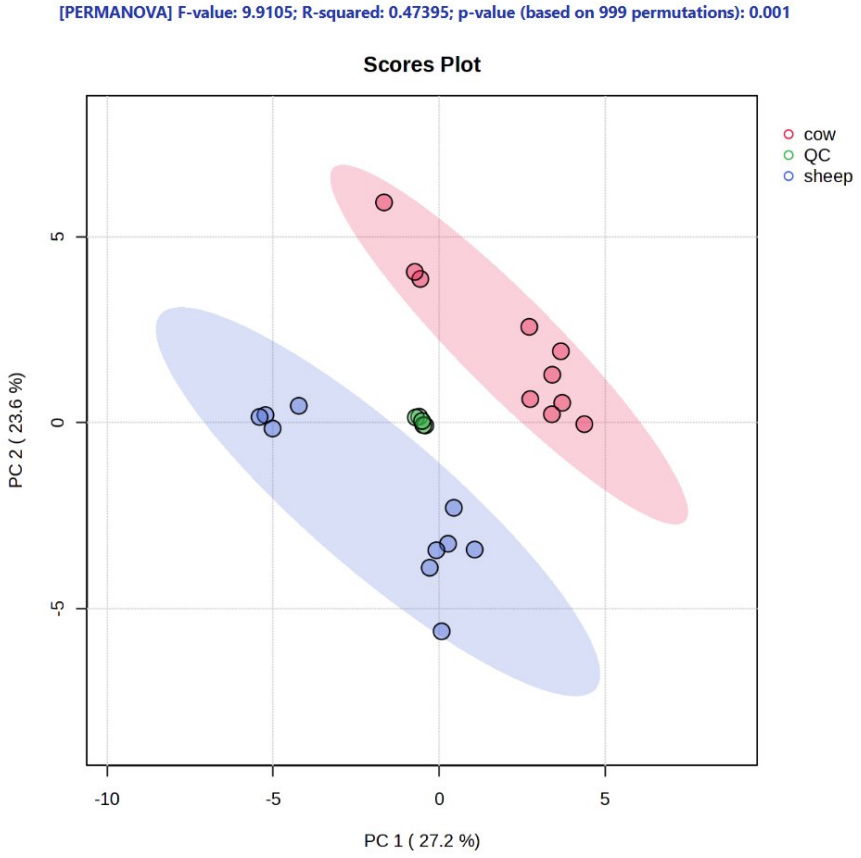


0.8 threshold: PQN vs sum normalisation

PQN



sum

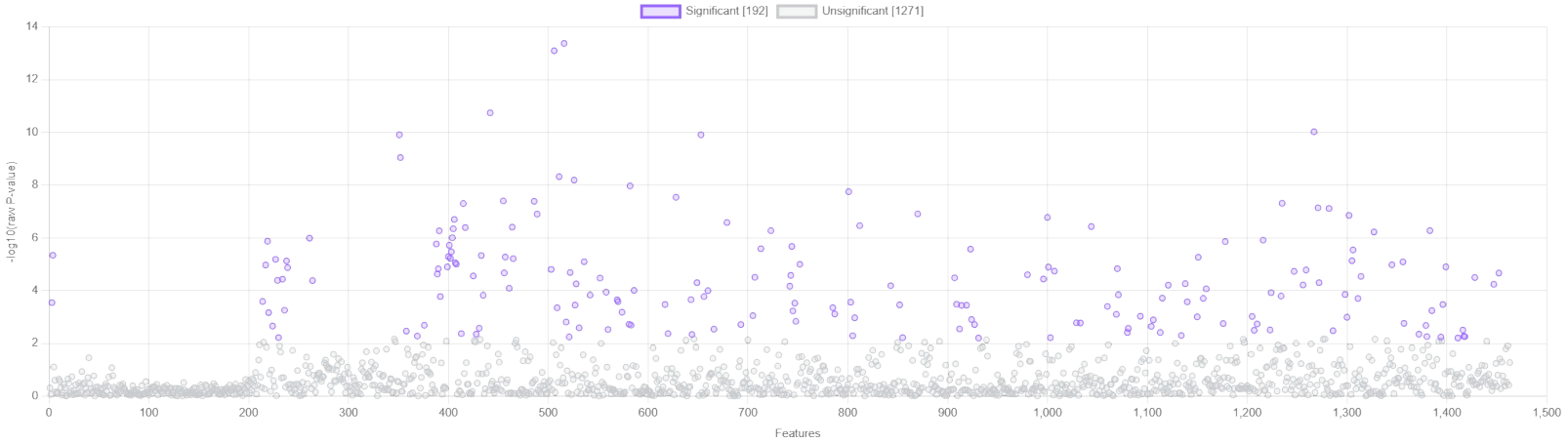


- PQN normalisation has a higher F-value, showing that it maximises the variance between the groups but minimises the variance within the groups.
- The R-value for PQN is higher, meaning it captures more variance within the data.
- PQN shows greater distinction between the cow and sheep groups as there is a larger variance.
- Based on this interpretation, further analyses will be focused around the 0.8 threshold PQN normalised data.

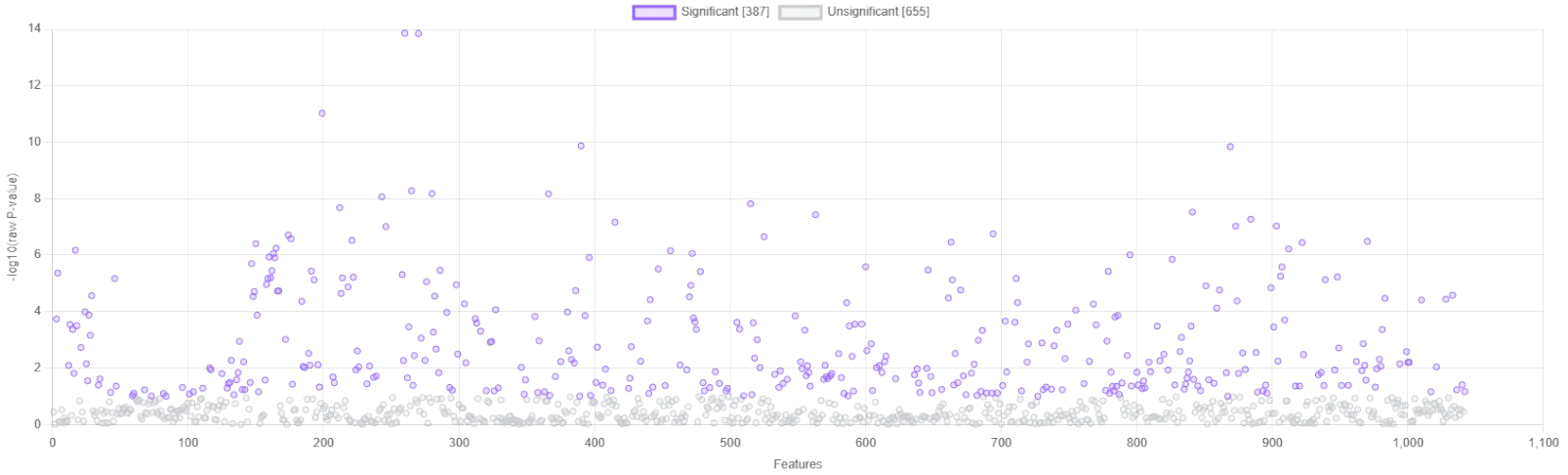
Comparison of 0.5 and 0.8 thresholds: Wilcoxon T-test

- As seen in the volcano plot, the overall feature count has decreased in the 0.8 threshold dataset.
- The 0.8 threshold uses stricter filters, so it contains more significant, biologically relevant features.
- Both datasets contain the same 2 highest features, showing repeatability of the processing and analysis.
- Overall, using 0.8 threshold data yields clearer and more accurate results.

0.5 threshold (P-value threshold (FDR)- 0.05)

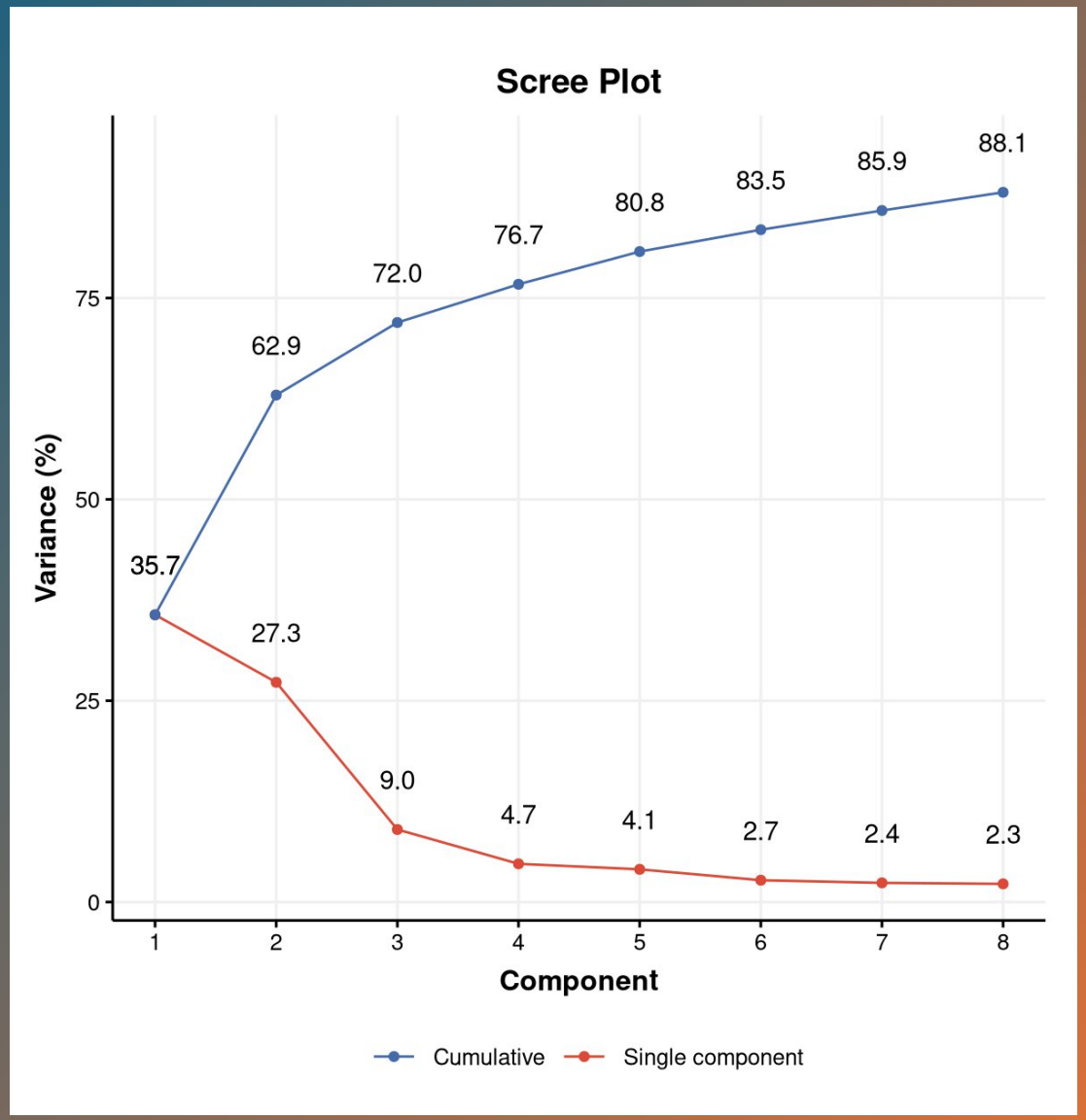


0.8 threshold (P-value threshold (FDR)- 0.05)



Workflow in R:

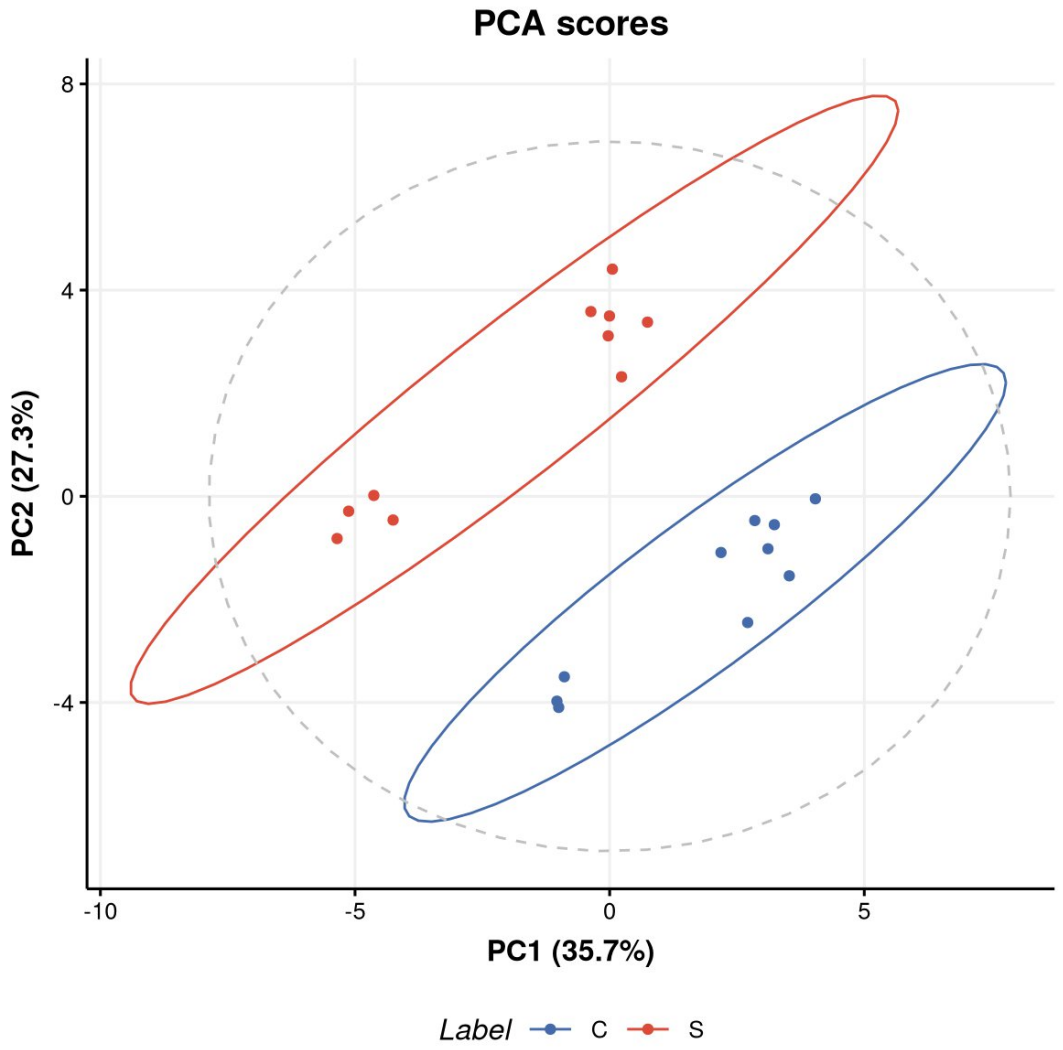
Scree Plot analysis



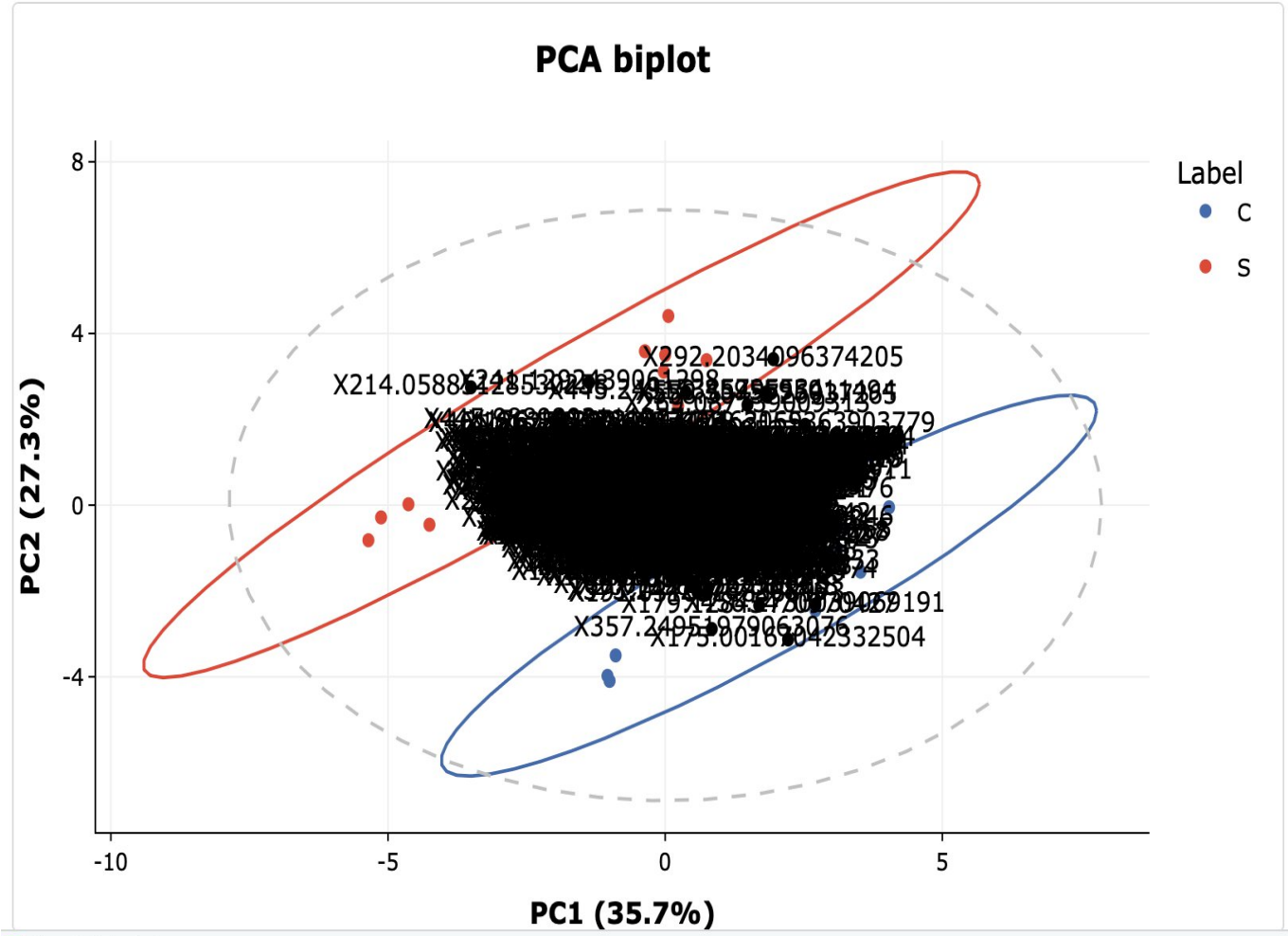
- PCA was performed to reduce dimensionality and summarise overall metabolic variation.
- **PC1 explains 35.7%** and **PC2 explains 27.3%** of the total variance.
- The first **5 principal components explain 80.8% cumulative variance**, exceeding the predefined threshold of **80% (0.8)**.
- Components beyond PC5 contribute <5% variance each and were not considered biologically significant.

PCA Scores

- Principal Component Analysis shows clear separation between **Control (C)** and **Sample (S)** groups.
- **PC1 explains 35.7%** and **PC2 explains 27.3%** of the total variance, together accounting for **63.0%** of data variation.
- Samples cluster according to their biological groups, indicating distinct metabolic profiles
- Minimal overlap between group confidence ellipses suggests good group discrimination and low within-group variability.
- The clustering pattern reflects consistent experimental conditions and reliable data quality.



PCA Biplot Analysis



The PCA biplot combines sample scores and metabolite loadings to visualise group separation and variable contributions.

PC1 (35.7%) and PC2 (27.3%) together explain 63.0% of the total variance.

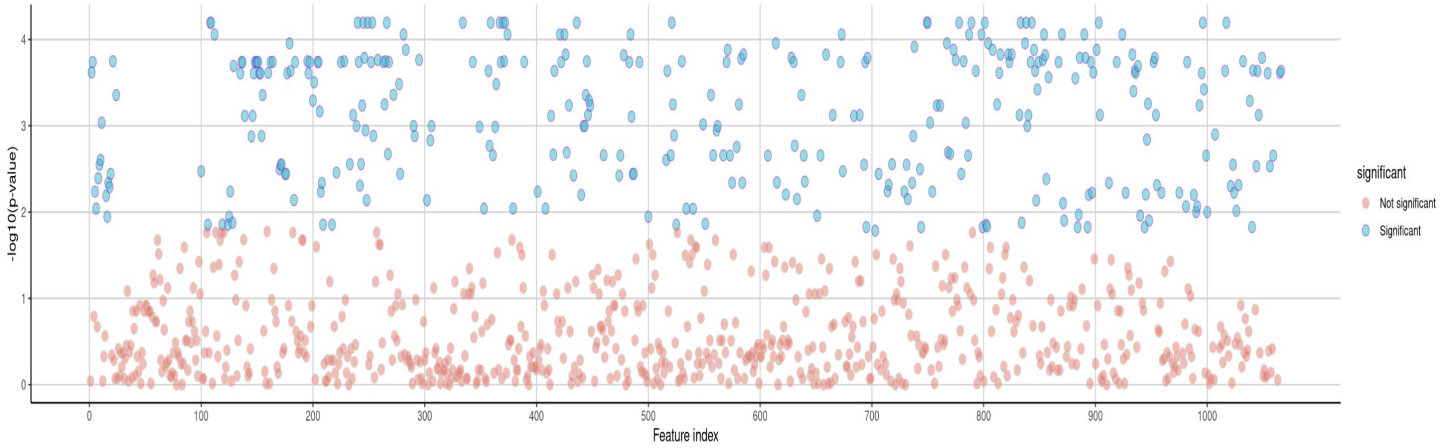
Clear separation is observed between **Control (C)** and **Sample (S)** groups, confirming distinct metabolic profiles.

Most metabolites cluster near the origin, indicating moderate contributions to overall variance.

A subset of metabolites shows stronger loadings along PC1 and PC2, suggesting their key role in driving group discrimination.

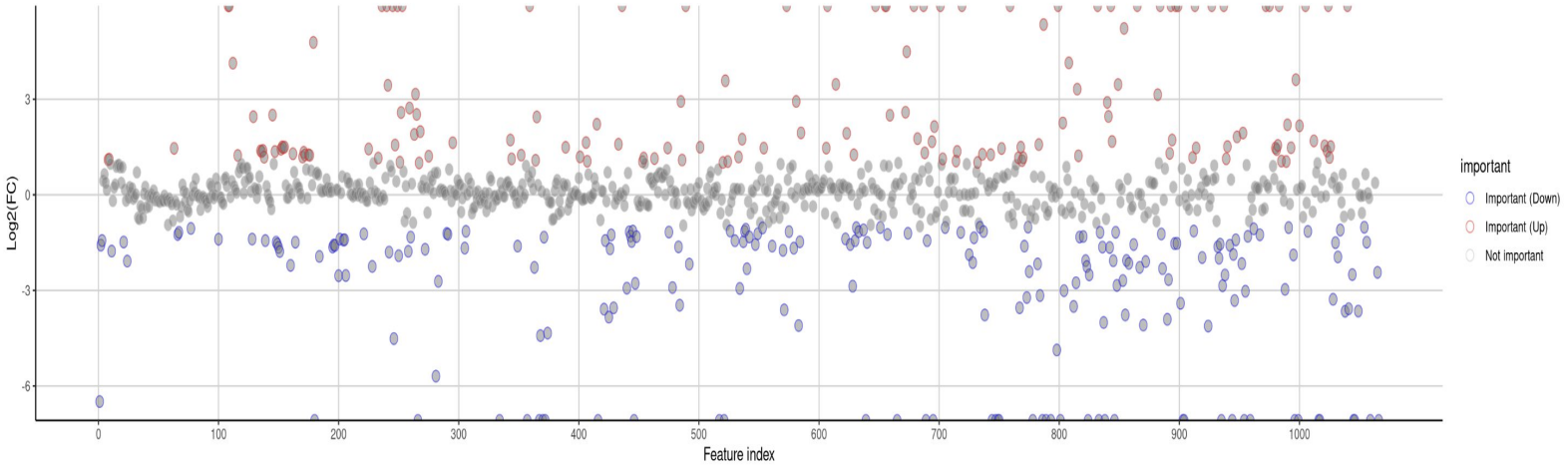
Overlapping feature labels reflect the high dimensionality typical of untargeted metabolomics datasets.

Nonparametric t-test



- The plot shows the statistical significance of metabolites based on **$-\log_{10}$(p-value)** from a non-parametric t-test.
- Each point represents an individual metabolite feature across the dataset.
- **Blue points** indicate statistically significant metabolites, while **red points** represent non-significant features.
- A large number of metabolites display high $-\log_{10}$(p-values), suggesting strong differences between groups.
- Most non-significant features cluster at low $-\log_{10}$(p-values), indicating minimal group variation.

Fold change



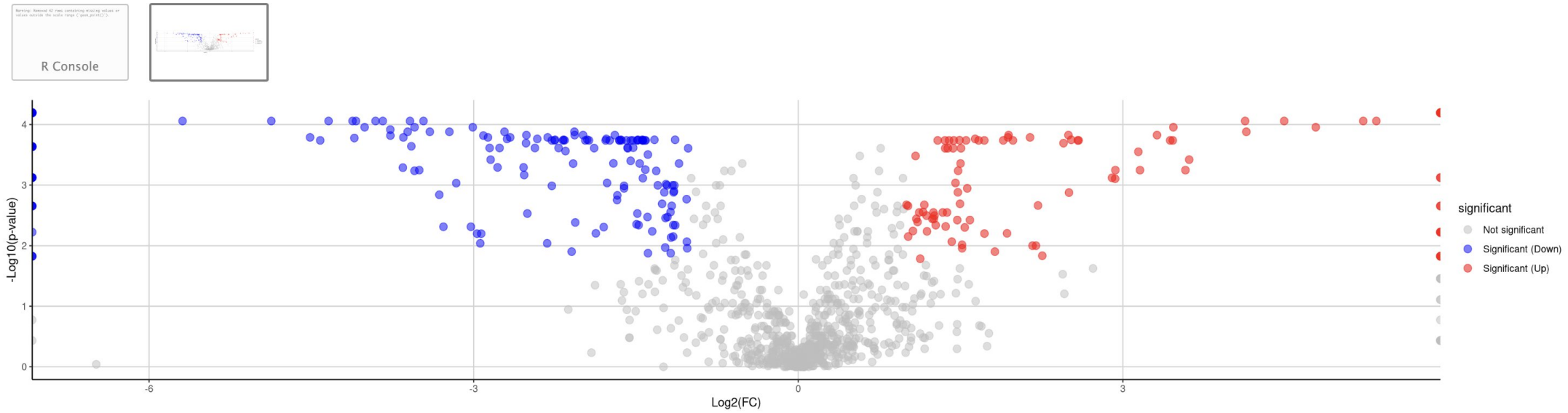
The plot shows changes in metabolite abundance between groups expressed as **$\log_2$ fold change (FC)**.

Red points represent significantly upregulated metabolites, while **blue points** indicate downregulated metabolites.

Most features cluster around **$\log_2$FC ≈ 0**, suggesting no major change between groups.

A subset of metabolites shows large positive or negative $\log_2$FC values, indicating strong differential expression.

Volcano Plot



- The volcano plot displays the relationship between **$\log_2$ fold change (FC)** and **statistical significance ($-\log_{10}$ p-value)**.
- The **x-axis ($\log_2$FC)** represents the magnitude and direction of change between groups.
- The **y-axis ($-\log_{10}$ p-value)** indicates statistical significance, with higher values showing stronger confidence.
- **Red points** represent significantly upregulated metabolites, while **blue points** indicate significantly downregulated metabolites.
- **Grey points** are not statistically significant and show minimal or inconsistent changes.
- Metabolites located in the upper left and upper right regions show both large fold changes and high significance.

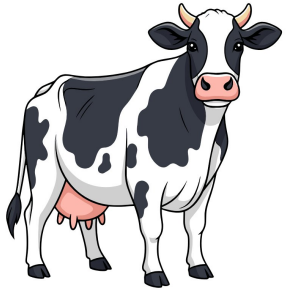
Annotation



Annotation Overview







Used Database The Human Metabolome Database 5.0
Kyoto Encylopedia of Genes and Genomes

Input Data:

The statistically significant m/z features derived from volcano analysis after preprocessing and normalisation.

Input Selection Criteria

- Fold change (FC) ≥ 2
- Therefore $\log_2$FC ≥ 1 (Cow/Sheep)
- p-value ≤ 0.1
- Data PQN-normalised (MetaboAnalyst)
- Sample filtering threshold: 0.8 minimum fraction (Galaxy)

hmdb The Human Metabolome Database

KEGG Kyoto Encyclopedia of Genes and Genomes

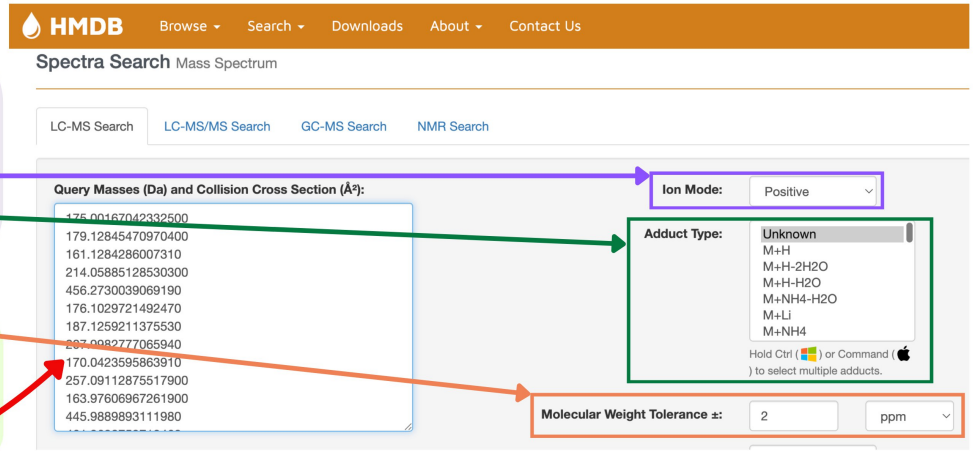
Input Data:

KEGG Compound ID for the chosen metabolite in HMDB.

Database Search Parameters

- **Ion mode:** Positive
- **Adduct type:** Unknown (all relevant positive-mode adducts considered)
- **Mass tolerance:** ≤ 2 ppm

Query Only the m/z values of those significant features



Generalised Annotation Criteria

SEPARATING SIGNAL FROM NOISE

PRECISION

Mass error tolerance (Δ, ppm)

Only features with mass error ≤ 2 ppm (relative to theoretical adduct m/z) were considered putative matches.

$$\Delta(ppm) = \left(\frac{|m/z_{observed} - m/z_{theoretical}|}{m/z_{theoretical}} \right) \times 10^6$$

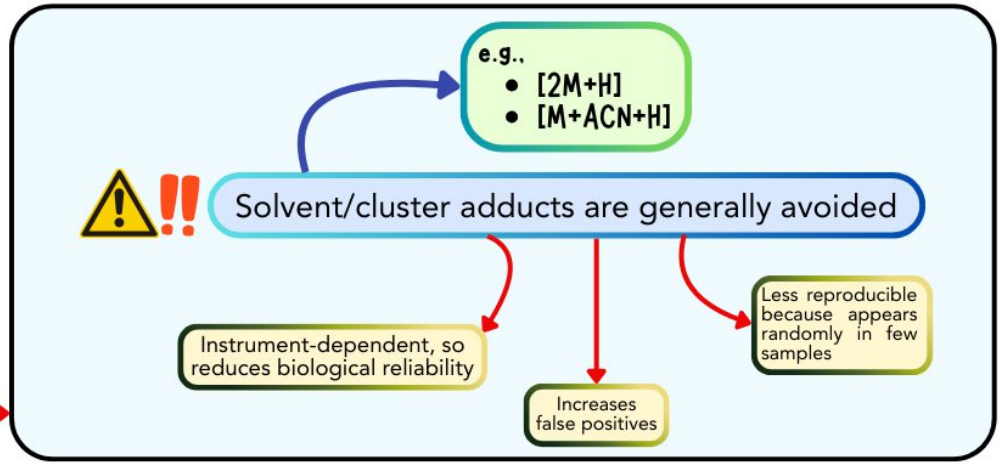
CHEMISTRY

Adduct Logic

Preference given to primary adducts like:

- M+H (protonated)
- M+Na (sodiated)
- M+K (potassiated)

Primary adducts form naturally in electrospray ionisation and are chemically expected.

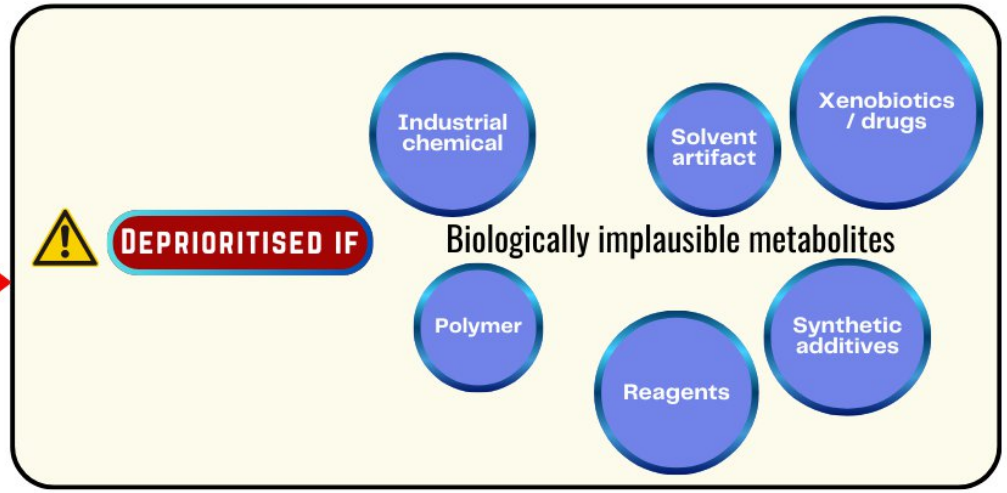


CONTEXT

Biological Plausibility

Putative annotations were prioritised based on the following biological context:

- Known presence in mammalian cardiac tissue
- Relevance to ruminant metabolism
- Plausibility in relation to: Diet, known biochemical pathways or endogenous metabolite.
- Abundant or widely recognised in literature



Other Followed Annotation Criteria



Grouping Multiple m/z Features by Neutral Mass

- No LC-MS or fragmentation (MS/MS)
- Same neutral metabolite ↔ multiple m/z features
- **Different adduct formations during ionisation.**

So,

- Features grouped by neutral mass
- Multiple ion single putative metabolite.

Citrulline (an alpha-amino acid) detected as:

- [M+H]+ (m/z 176.1030)
- [M+Na]+ (m/z 198.0849)
- [M+K]+ (m/z 214.0589)

EXAMPLE



Handling of Isomers

- MS1-only DIMS data cannot distinguish isomers
- Multiple compounds = same (molecular formula + m/z)
- **Biologically abundant and commonly reported one chosen**

$C_5H_9NO_4$ isomers at m/z 170.0424 were:

- O-acetylserine
- N-methyl-aspartate
- Hydroxy-oxo-norvaline
- Glutamic acid

EXAMPLE

Glutamate was most abundant, so it was chosen for that m/z value



Lipid Annotation Rules

- Class-level annotation when biologically plausible
- Specific sn-positions
- So structural ambiguity
- Annotation done based on:
  - commonly reported in animal tissue
  - adduct type

PA(22:6/18:2) at m/z 373.2442 chosen despite other PAs with same formula and adduct:

- PA(20:5/20:3)
- PA(22:5/18:3)
- PA(22:4/18:4), etc.

EXAMPLE

DHA- and linoleate-containing phospholipids common in animal tissues.

Applied Basic Annotation Confidence



HIGH CONFIDENCE

- Δ 0-1 ppm
- Primary adduct(s): [M+H]+, [M+Na]+, [M+K]+
- **Adduct Grouping**
- Strong cardiac plausibility
- **No** conflicting **isomers**

- **Hypoxanthine** EXAMPLE
- m/z: 175.0017 as [M+K]+ and 159.0277 as [M+Na]+
- Core purine metabolism intermediate
- Common in mammalian tissues



MEDIUM CONFIDENCE

- Δ ≤ 2 ppm
- No clear primary adduct(s): [M+NH4]+,[M+2H]+, [M+2K-H]+, etc.
- Often biologically plausible
- Primary adduct but isomer ambiguity
- Dietary exposure, no direct tissue-specific support
- No Xenobiotic match

- **Flavin mononucleotide (FMN)** EXAMPLE
- m/z: 533.0226
- Core endogenous cofactor used in ETC.
- [M+2K-H]+ adduct



LOW CONFIDENCE

- Δ ≤ 2 ppm
- Solvent/cluster adduct(s): [M + **ACN/DMSO/IsoProp/ H2O** + H]+
- Weak biological context
- Multiple unresolved isomers with no prioritisation basis

- **4-Phenylbutyl glucosinolate**
- m/z: 416.0843 EXAMPLE
- Plant-derived metabolite (from Brassica/forage)
- [M+H-2H2O]+ adduct



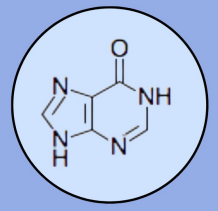
VERY LOW CONFIDENCE

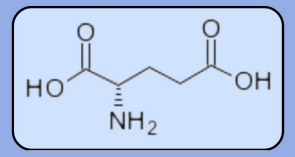
- **Δ ≤ 2 ppm**
- **Biologically implausible**
- **Only solvent/cluster adducts**
- **No mammalian cardiac relevance**

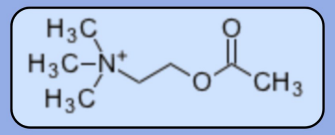
- **1,1-Bis(p-chlorophenyl)-2-chloroethene**
- m/z: 326.9483
- Pesticide
- biologically implausible
- [M+2Na-H]+ adduct

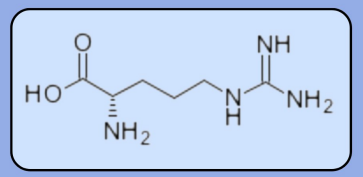
Key Annotation Findings

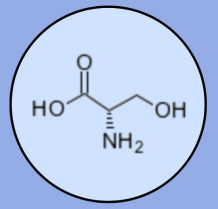
Enriched in Cow


Hypoxanthine


Glutamic acid


Acetylcholine


L-Arginine


Serine

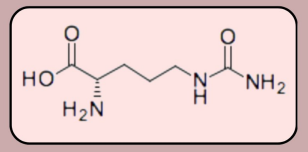
Purine metabolism

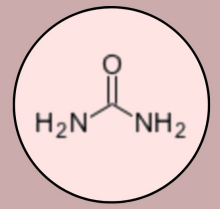
Amino acid metabolism

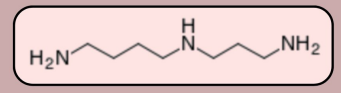
TCA cycle

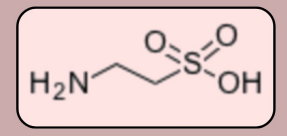
Cholinergic pathway

Enriched in Sheep


Citrulline


Urea


Spermidine


Taurine

Amino acid metabolism

Urea Cycle

Arginine Biosynthesis

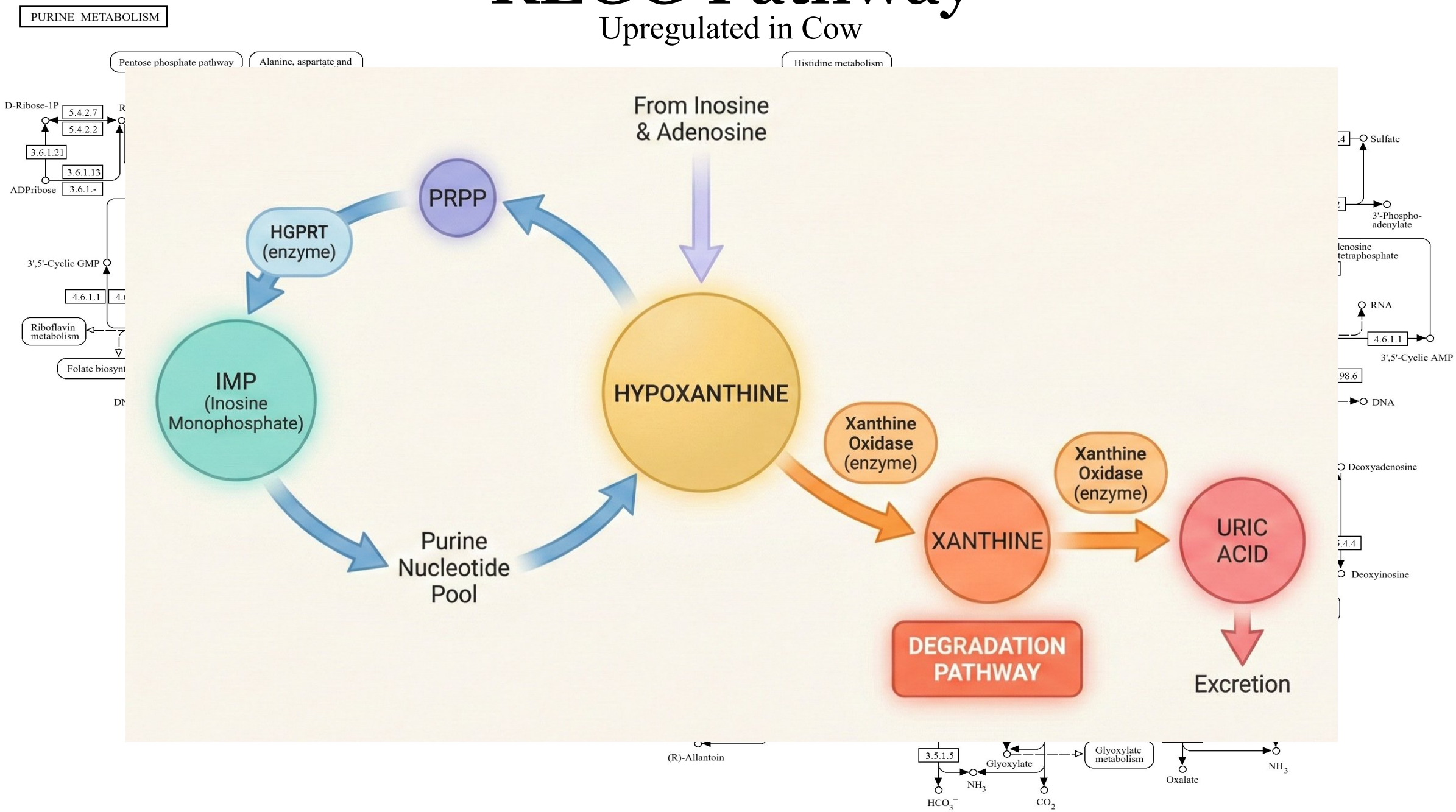
Taurine metabolism

Spermidine and spermine biosynthesis

COMPLETED

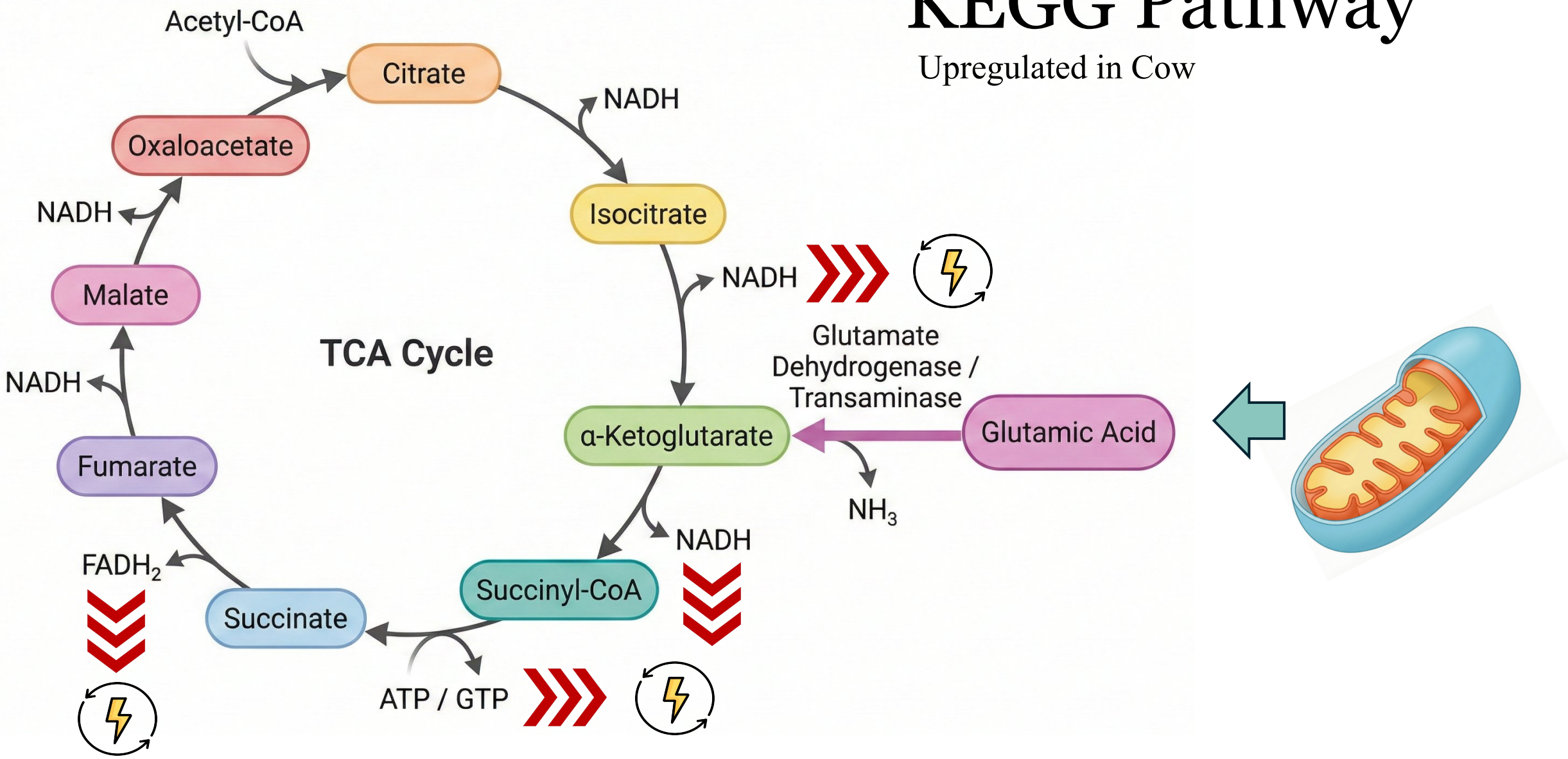
KEGG Pathway

Upregulated in Cow



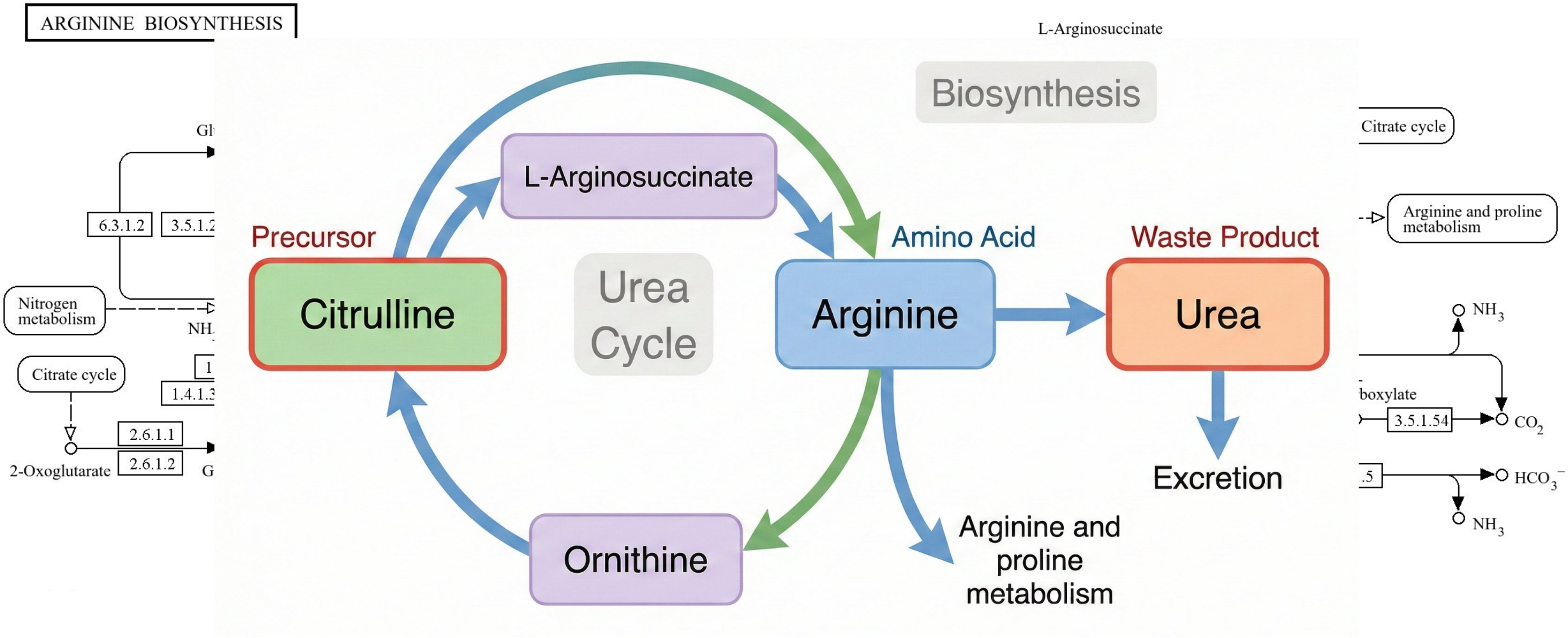
KEGG Pathway

Upregulated in Cow



KEGG Pathway

Upregulated in Sheep



Group contributions

Ranveer Basra

- TSV data cleaning (replacing 0's with blanks and converting TSV to CSV).
- Metaboanalyst processing and analysis:
  - Pipeline creation
  - Optimisation of parameters for the datasets.
  - Statistical analysis with plot generation (PCA, Volcano and Wilcoxon) and model statistics.
  - Comparison of normalisation (PQN vs Sum) and datasets (0.5 threshold vs 0.8 threshold).
- Slide preparation with pipeline, plots and analysis.

Shalini Majumder

- Data Exploration (of filtering parameter changes) using Python
- Putative metabolite annotation using HMDB and KEGG:
  - Annotation criteria determination
  - Identifying key metabolites and their associated metabolic pathways
- Main work pipeline preparation
- Introductory, data exploration, and annotation slides preparation

Hattim Mohammad

- Data preprocessing using Galaxy
  - Pipeline justification
  - Tweaking and optimising parameters
  - Statistical analysis in Galaxy
  - Comparison of sample filters
- Exploratory data analysis in R
- Slide preparation with pipeline, plots and analysis.

Divya Kadrekar

- R structToolbox analysis:
  - Performed PCA and scree plot analysis for dimensionality reduction
  - Identified optimal number of principal components (>80% variance)
  - Analysed group separation using PCA score plots
  - Interpreted PCA biplot to identify influential metabolites
  - Conducted $\log_2$ fold change analysis
  - Generated and analysed volcano plots for differential expression
  - Applied non-parametric t-test for significance testing

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- David S Wishart, AnChi Guo, Eponine Oler, Fei Wang, Afia Anjum, Harrison Peters, Raynard Dizon, Zinat Sayeeda, Siyang Tian, Brian L Lee, Mark Berjanskii, Robert Mah, Mai Yamamoto, Juan Jovel, Claudia Torres-Calzada, Mickel Hiebert-Giesbrecht, Vicki W Lui, Dorna Varshavi, Dorsa Varshavi, Dana Allen, David Arndt, Nitya Khetarpal, Aadhavya Sivakumaran, Karxena Harford, Selena Sanford, Kristen Yee, Xuan Cao, Zachary Budinski, Jaanus Liigand, Lun Zhang, Jiamin Zheng, Rupasri Mandal, Naama Karu, Maija Dambrova, Helgi B Schiöth, Russell Greiner, Vasuk Gautam, HMDB 5.0: the Human Metabolome Database for 2022, *Nucleic Acids Research*, Volume 50, Issue D1, 7 January 2022, Pages D622–D631, https://doi.org/10.1093/nar/gkab1062

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Thank You.

Questions & Discussion

